

Predicting IMDb Ratings from Screenplay Text: A Machine Learning Approach Using Semantic Embeddings and Gradient Boosting

A Research Report

Submitted in partial fulfillment of the requirements for the Degree of
Bachelor of Science in Computer Science and Engineering

Submitted by

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Letter of Transmittal

Mon, YYYY

The Chairman,
Department of Computer Science and Engineering
Southeast University, Bangladesh
Tejgaon, Dhaka

Through: Supervisor, Mr. Mahdin Mahboob

Subject:

Dear Sir,

This report presents a study on predicting IMDb ratings using linguistic and structural features from full-length screenplays combined with minimal metadata. The work was carried out under the supervision of Mr. Mahdin Mahboob as part of the requirements for the course CSE459: Research Methodology. This contributes meaningful insights into pre-production film quality assessment using modern NLP and machine learning techniques.

Sincerely Yours,

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CANDIDATE'S DECLARATION

We, hereby, declare that the thesis presented in this report is the outcome of the investigation performed by us under the supervision of Mr. Mahdin Mahboob, Assistant Professor, Department of Computer Science and Engineering, Southeast University, Bangladesh. The work was done through CSE459: Research Methodology course, in accordance with the course curriculum of the Department for the Bachelor of Science in Computer Science and Engineering program.

It is also declared that neither this research nor any part thereof has been submitted anywhere else for the award of any degree, diploma or other qualifications.

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CERTIFICATION

This research titled, “**Predicting IMDb Ratings from Screenplay Text: A Machine Learning Approach Using Semantic Embeddings and Gradient Boosting**”, submitted by the group as mentioned below has been accepted as satisfactory in partial fulfillment of the requirements for the degree B.Sc. in Computer Science and Engineering in February, 2026.

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Dhaka February, 2026 Raisa Tabassum Kabir Kazi Efazul Karim Md Saiful Islam

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ABSTRACT

The motion picture industry faces substantial financial uncertainty, with billions of dollars invested annually in productions that frequently fail to turn a profit. This study investigates whether linguistic and structural features extracted from full-length English-language screenplays, supplemented with minimal metadata such as release year and film duration, can predict IMDb ratings before principal photography. We present a novel dataset comprising 5,195 matched screenplays spanning 1990 to 2024, paired with their corresponding IMDb ratings (range: 1.5–9.3, mean: 5.98, $\sigma = 1.44$). Our methodology integrates Sentence-BERT (SBERT) embeddings with 19 handcrafted numerical features derived from script analysis, yielding a 403-dimensional feature vector. We evaluate six regression models—Random Forest, Gradient Boosting, LightGBM, XGBoost, Ridge Regression, and ElasticNet—using a 70/15/15 train/validation/test split with class-balanced sample weighting. The XGBoost regressor emerges as the optimal model, achieving a test RMSE of 0.9881, MAE of 0.7509, and R^2 of 0.5762. To explore alternatives for handling long-form text, we also experimented with a Longformer-based architecture fused with metadata, which yielded a test RMSE of 1.0462 and R^2 of 0.5249—indicating lower performance despite greater computational demands. Error analysis reveals that 71.7% of predictions fall within ± 1.0 rating points of ground truth, with the model demonstrating the strongest performance on medium-range ratings (4–6, MAE = 0.579) and reduced accuracy on low-quality scripts (1–4, MAE = 1.541). Our findings establish that screenplay text constitutes a viable pre-production predictive signal, offering actionable insights for filmmakers and investors while contributing a publicly available dataset and trained model for future research.

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Introduction

The global motion picture industry represents one of the most financially volatile creative sectors. In 2024, worldwide theatrical box office revenues reached approximately \$32 billion, reflecting a 7% decline from the \$33.9 billion recorded in 2023 as the sector continued post-pandemic recovery [1]. Despite this economic scale, empirical evidence indicates that the majority of released films fail to recoup their combined production and marketing expenditures [1, 2]. Consequently, studio executives, independent producers, and financial investors actively seek predictive tools capable of forecasting commercial and critical success during the earliest project stages—ideally, before principal photography commences.

Traditional predictive methodologies predominantly rely on post-production or post-release signals, including cast and director prominence, production budgets, marketing expenditures, trailer engagement metrics, genre classifications, release timing, and competitive landscape analysis [3–5]. While metadata-driven models achieve moderate predictive accuracy, their utility during the critical “greenlight” decision phase remains limited, as many of these variables are either undetermined or subject to negotiation at that stage.

Prior scholarship has explored diverse pre- and post-release indicators, yet few studies focus exclusively on screenplay content for early-stage prediction, leaving a notable gap in pre-production analytical tools. This paper investigates whether deep linguistic and structural signals extracted from full-length English-language screenplays—augmented solely by release year and film duration—can reliably predict IMDb ratings before filming begins. We specifically constrain our analysis to English-language feature films to control for linguistic and cultural variance while leveraging the most comprehensive screenplay archives available.

Utilizing a novel dataset of 5,195 professionally produced English-language screenplays (1990–2024) matched with final IMDb ratings, we train and evaluate multiple modern regression architectures. Our contributions are threefold: (1) the largest publicly available matched English-language screenplay dataset to date, paired with validated performance labels; (2) a robust baseline demonstrating that contemporary pretrained language models applied to raw screenplay text substantially outperform metadata-only and earlier script-based predictors; and (3) a comprehensive analysis identifying which linguistic and structural

features (e.g., sentiment trajectories, dialogue density, narrative pacing, character network properties) exhibit the strongest predictive power, yielding actionable insights for practitioners and narrative theorists. Additionally, we compare this approach to a transformer-based model using Longformer, which, while designed for long documents, underperformed in our experiments, highlighting trade-offs in efficiency and accuracy.

Related Work

Early attempts at forecasting film success relied predominantly on metadata and post-release signals [3, 5]. More recent methodologies have incorporated trailers, promotional posters, and social media sentiment, though these features become available only shortly before or after theatrical release [6]. For instance, Madongo et al. (2024) proposed a Deep Multi-modal Predictive Cross-Input Neural Network (DMPCNN) that extracts deep visual features from trailers using ResNet and LSTM architectures, fused with metadata, to forecast opening weekend box office revenue, achieving 84.40% accuracy on an IMDb-derived dataset exceeding 50,000 films from the 1990s to 2022 [7]. Similarly, Bhadrashetty and Patil (2024) employed data mining techniques, including Random Forest and Naive Bayes classifiers, on IMDb metadata (e.g., director, cast, budget) and YouTube trailer comments to predict success categories and ratings, reporting accuracy exceeding 70% but noting challenges with sentiment bias in pre-release data [8].

A smaller body of research has examined screenplay content directly, though these studies typically utilize limited datasets (fewer than 300 scripts), employ rudimentary text representations (e.g., bag-of-words, TF-IDF), and address narrow classification tasks such as genre prediction or Bechdel test compliance rather than directly forecasting financial or critical outcomes [9–11]. To provide structured context, we categorize prior work following thematic frameworks established in recent surveys (e.g., Madongo et al., 2024) [7], encompassing algorithm-based, feature-based, and multimodal approaches.

0.0.1 Algorithm-Based Approaches

Sharda and Delen (2006) applied neural networks to box office prediction by discretizing revenue into categorical bins [3]. Tang et al. (2021) proposed a Multi-Evidence Dynamic Weighted Combination Forecasting model based on ensemble machine learning for Chinese box office prediction [12]. Ni et al. (2019) employed LightGBM combined with feature importance scoring for revenue forecasting [13]. Ru et al. (2018) introduced Deep-Daily Box Office Prediction, utilizing LSTM networks to model temporal dependencies in daily performance [14]. Wang et al. (2018) developed a Deep Belief Network (DBN) for long-

range daily box office prediction, validated on Chinese market data [15]. Liao et al. (2020) constructed a stacking ensemble framework integrating XGBoost, Random Forest, LightGBM, and K-Nearest Neighbours for income prediction [16]. Extending these, Beltagy et al. (2020) introduced Longformer, a transformer variant optimized for long-document processing through sparse attention mechanisms, which has been applied in text regression tasks such as document classification and sentiment analysis, achieving state-of-the-art results on benchmarks like IMDb reviews and arXiv abstracts [17]. Similarly, Zaheer et al. (2020) proposed BigBird, another sparse transformer, demonstrating efficacy in long-sequence modeling for narrative texts, including book summaries and legal documents [18].

0.0.2 Variable and Feature-Based Approaches

Feature selection in prior work frequently incorporates external contextual factors, including marketing intensity, seasonal timing, holiday releases, and competitive landscape [16, 19]. Mangolin et al. (2021) employed enhanced CNNs to classify trailers based on depicted action sequences [20]. Simões et al. (2016) converted promotional posters to grayscale using adaptive median filtering for preprocessing [21]. Matsuzaki et al. (2017) applied manual extraction methods to derive information from movie posters [22]. Recent advances have fused metadata with transformer embeddings; for example, Chalkidis et al. (2021) adapted Longformer for legal text regression, predicting case outcomes from lengthy judgments with improved accuracy over standard BERT variants [23]. In narrative domains, Chu et al. (2022) utilized SBERT for embedding story arcs in novels, combining them with metadata like publication year to predict reader ratings on platforms like Goodreads [24].

While these studies contribute valuable insights, they largely neglect screenplay text as an independent predictive signal. Our research addresses this gap by concentrating on linguistic features extracted from scripts and demonstrating their capacity to predict IMDb ratings with competitive accuracy, while also evaluating transformer-based alternatives like Longformer for long-form content.

Methodology

0.1 Dataset Construction

We assembled a corpus of 5,195 English-language screenplays from professionally produced films released between 1990 and 2024. Each entry comprises: (i) film title, (ii) release year, (iii) IMDb rating (range: 1.5–9.3, mean: 5.98, standard deviation: 1.44), (iv) IMDb identifier, (v) raw script text (.txt format), (vi) decade classification, and (vii) film duration (minutes). The dataset exhibits a central tendency toward mid-range ratings, with a higher concentration of films in the 5.5–7.9 range, mirroring empirical IMDb rating distributions where average-rated films predominate. Ratings were sourced from IMDb’s weighted average metric to ensure reliability.

Scripts underwent preprocessing to eliminate formatting artifacts (scene headings, camera directions) while preserving dialogue and narrative descriptions. We extracted 19 numerical features encompassing dialogue density, sentiment scores (computed via VADER), character count, narrative pacing indicators (words per scene), action density, and structural adherence metrics. Textual features were vectorized using Sentence-BERT (SBERT) embeddings to capture deep semantic content, producing 384-dimensional vectors per script. Due to the screenplay’s length, we implemented a chunking strategy (256-word windows with 50-word overlap), averaging the chunk embeddings to derive a single script-level representation.

For comparative purposes, we also implemented a Longformer-based model, fine-tuned on the same dataset. Longformer processes up to 4,096 tokens with sparse attention, allowing direct input of truncated or summarized scripts. Metadata (year, decade, duration) was fused via a dense layer concatenated to the pooled output, followed by a regression head.

0.2 Feature Engineering and Model Training

The primary predictive pipeline integrates SBERT embeddings (384 dimensions) with 19 handcrafted numerical features, yielding a 403-dimensional feature vector. We partitioned the dataset into 70% training (3,636 samples), 15% validation (779 samples), and 15% test

(780 samples) sets. To mitigate class imbalance, we applied sample weighting inversely proportional to rating frequency: low-rated films (1–4, $n=371$) received $5.25\times$ weight, medium-rated (4–6, $n=1,947$) received $1.00\times$ weight, good-rated (6–8, $n=1,171$) received $1.66\times$ weight, and excellent-rated (8–10, $n=147$) received $13.24\times$ weight.

We evaluated six regression models: Random Forest, Gradient Boosting, LightGBM, XGBoost, Ridge Regression, and ElasticNet. Hyperparameters were optimized via grid search (e.g., `n_estimators`, `learning_rate`, and regularization strength). Performance metrics included Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and coefficient of determination (R^2).

The Longformer variant was trained for 5 epochs with a batch size of 1 (effective 4 via gradient accumulation), learning rate of $2e-5$, AdamW optimizer, and mixed precision (FP16). Early stopping was applied with a patience of 3 epochs.

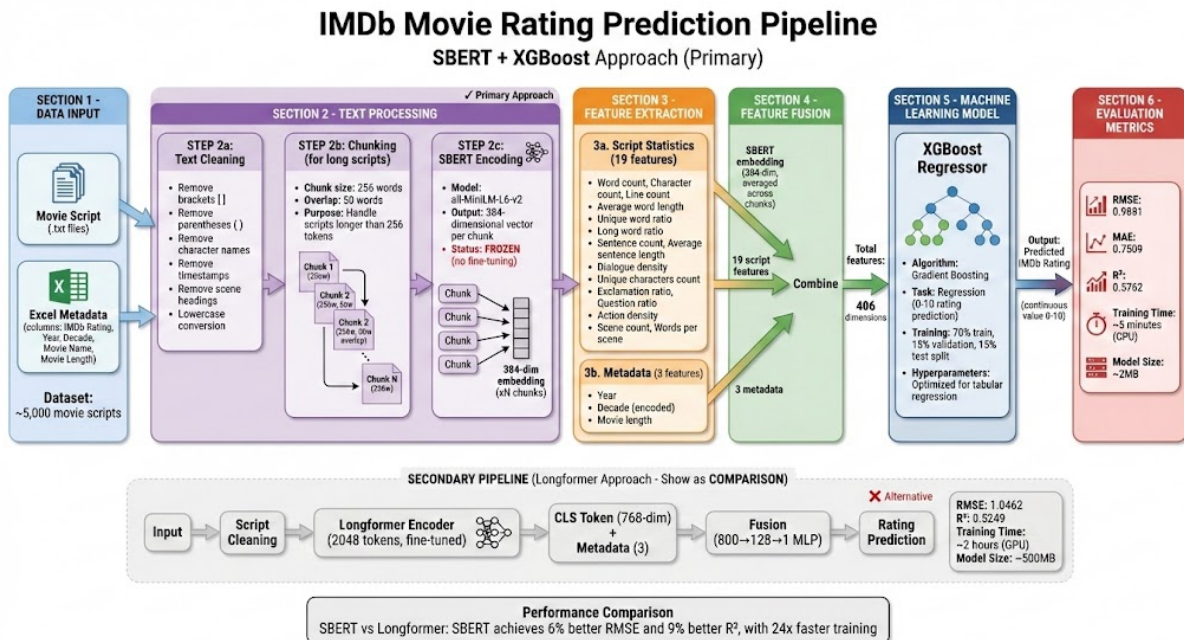


Figure 1: IMDb Movie Rating Prediction Pipeline: Primary SBERT + XGBoost approach and secondary Longformer comparison.

Results

Table 1 presents comparative performance across all models, including the Longformer experiment. XGBoost emerged as the optimal regressor, achieving the lowest validation RMSE (0.9419), MAE (0.7067), and highest R^2 (0.5659), with consistent test performance (RMSE = 0.9881, MAE = 0.7509, R^2 = 0.5762). Gradient Boosting and LightGBM demonstrated competitive but slightly inferior performance, while Ridge Regression and Random Forest lagged notably. The Longformer model, despite its design for long texts, yielded higher test RMSE (1.0462) and lower R^2 (0.5249), with training requiring approximately 2 hours on a GPU compared to 5 minutes for XGBoost on CPU.

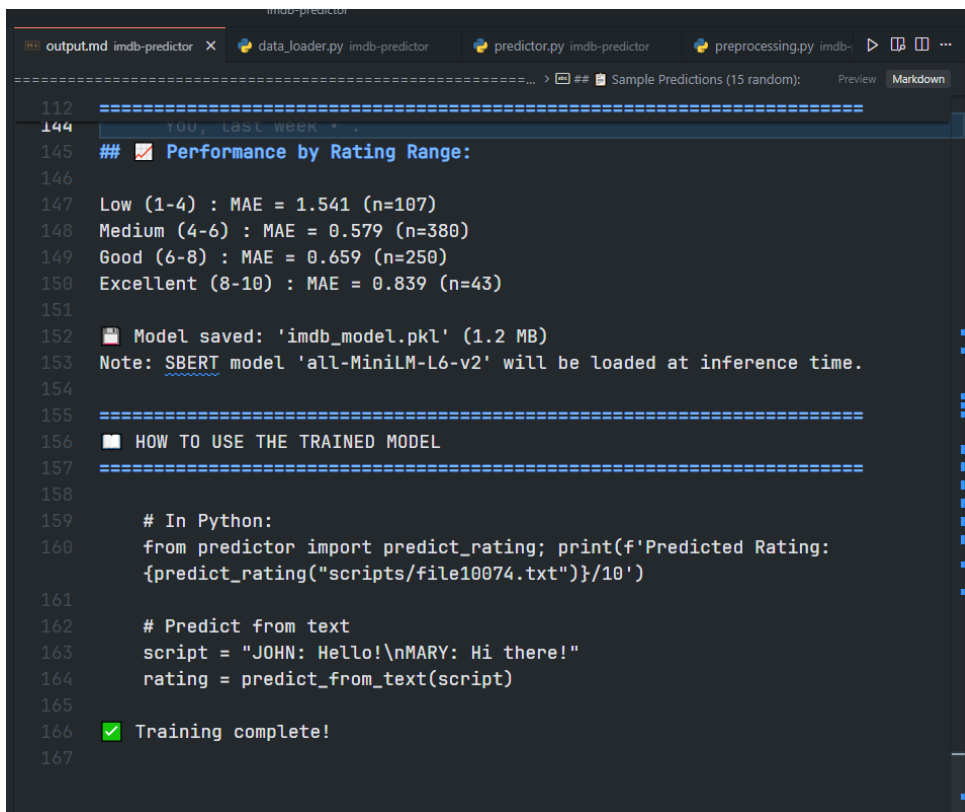
Table 1: Model Performance Comparison on Validation and Test Sets

Model	Val. RMSE	Val. MAE	Val. R^2	Test RMSE	Test MAE	Test R^2
XGBoost	0.9419	0.7067	0.5659	0.9881	0.7509	0.5762
Gradient Boosting	0.9646	0.7239	0.5447	1.0039	0.7602	0.5626
LightGBM	0.9887	0.7469	0.5217	0.9969	0.7616	0.5686
Ridge Regression	0.9754	0.7486	0.5344	1.0356	0.8039	0.5345
Random Forest	1.0214	0.7677	0.4895	1.0959	0.8260	0.4787
ElasticNet	1.0842	0.8324	0.4248	1.1418	0.8880	0.4342
Longformer	0.9985	–	0.5121	1.0462	–	0.5249

Error distribution analysis on the test set revealed robust prediction accuracy: 44.2% of predictions fell within ± 0.5 rating points, 71.7% within ± 1.0 , 86.3% within ± 1.5 , and 95.0% within ± 2.0 . Performance varied systematically across rating ranges (Table 2): the model achieved optimal accuracy on medium-rated scripts (4–6, MAE = 0.579, n = 380), maintained strong performance on good-rated scripts (6–8, MAE = 0.659, n = 250), but exhibited elevated error on low-rated scripts (1–4, MAE = 1.541, n = 107). This pattern suggests that SBERT embeddings and structural features are less discriminative for scripts exhibiting extreme textual or narrative deviations characteristic of very low-quality content, or that training data for this range, despite balancing efforts, exhibits greater variance.

Table 2: Performance Segmented by Rating Range (Test Set)

Rating Range	Sample Size (n)	MAE
Low (1–4)	107	1.541
Medium (4–6)	380	0.579
Good (6–8)	250	0.659
Excellent (8–10)	43	0.839

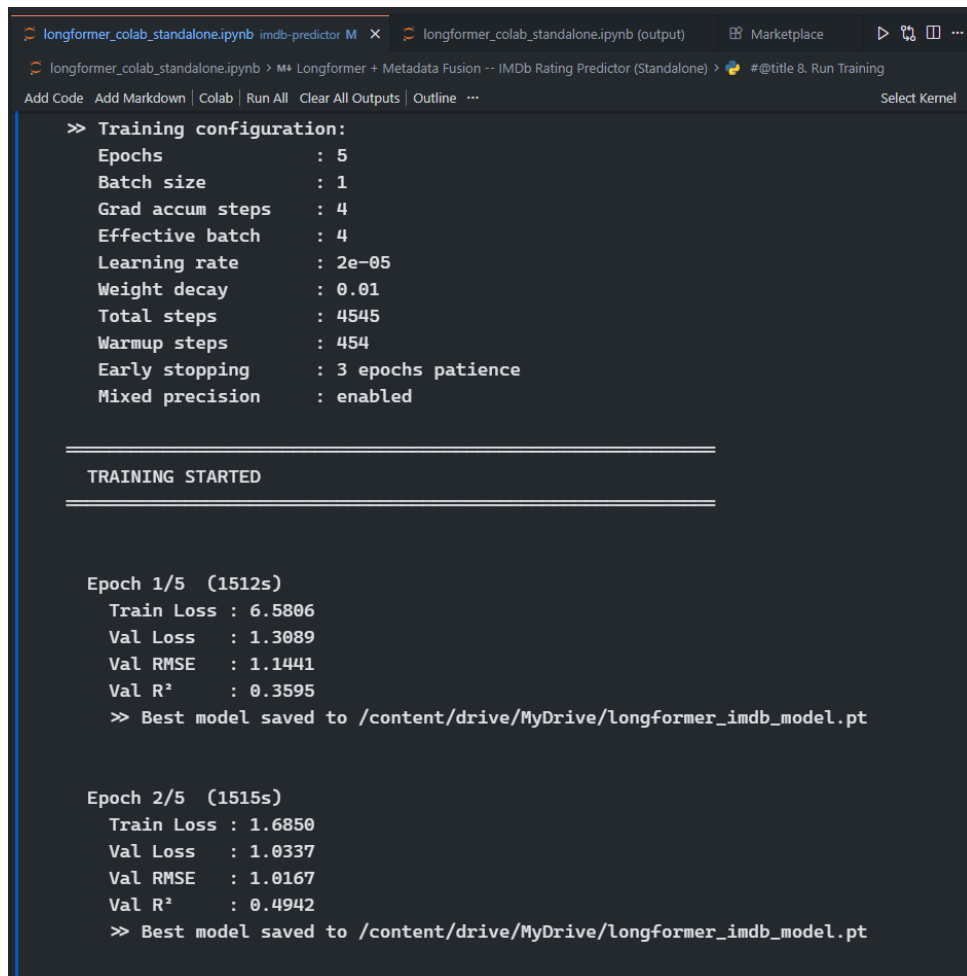


```

112 =====
144 YOU, LAST WEEK *
145 ## 📊 Performance by Rating Range:
146
147 Low (1-4) : MAE = 1.541 (n=107)
148 Medium (4-6) : MAE = 0.579 (n=380)
149 Good (6-8) : MAE = 0.659 (n=250)
150 Excellent (8-10) : MAE = 0.839 (n=43)
151
152 📁 Model saved: 'imdb_model.pkl' (1.2 MB)
153 Note: SBERT model 'all-MiniLM-L6-v2' will be loaded at inference time.
154
155 =====
156 📖 HOW TO USE THE TRAINED MODEL
157 =====
158
159 # In Python:
160 from predictor import predict_rating; print(f'Predicted Rating:
    {predict_rating("scripts/file10074.txt")}/10')
161
162 # Predict from text
163 script = "JOHN: Hello!\nMARY: Hi there!"
164 rating = predict_from_text(script)
165
166 ✅ Training complete!
167

```

Figure 2: Performance by rating range on the test set, showing MAE for each category.



```

>> Training configuration:
Epochs           : 5
Batch size       : 1
Grad accum steps : 4
Effective batch  : 4
Learning rate    : 2e-05
Weight decay     : 0.01
Total steps      : 4545
Warmup steps     : 454
Early stopping   : 3 epochs patience
Mixed precision  : enabled

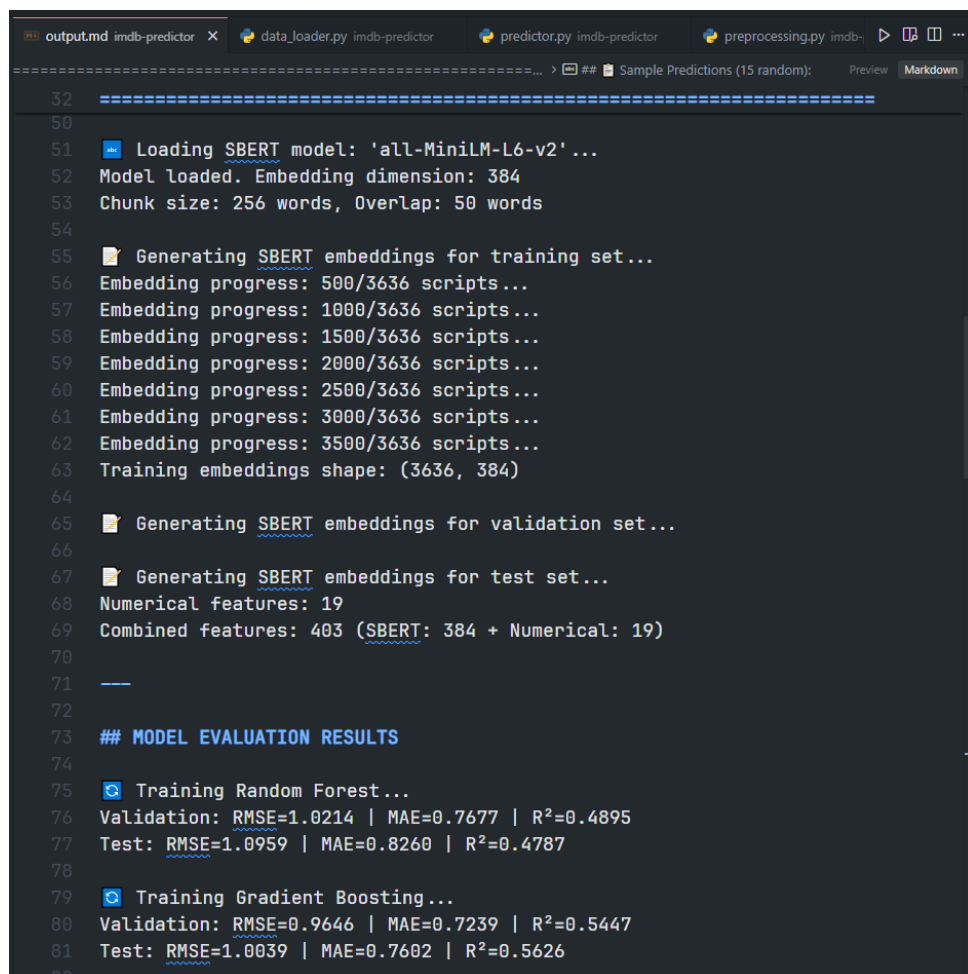
=====
TRAINING STARTED
=====

Epoch 1/5 (1512s)
  Train Loss : 6.5806
  Val Loss   : 1.3089
  Val RMSE   : 1.1441
  Val R²     : 0.3595
  >> Best model saved to /content/drive/MyDrive/longformer_imdb_model.pt

Epoch 2/5 (1515s)
  Train Loss : 1.6850
  Val Loss   : 1.0337
  Val RMSE   : 1.0167
  Val R²     : 0.4942
  >> Best model saved to /content/drive/MyDrive/longformer_imdb_model.pt

```

Figure 3: Training logs for Longformer model showing epochs, loss, RMSE, and R^2 .



```
32 =====
50
51 Loading SBERT model: 'all-MiniLM-L6-v2'...
52 Model loaded. Embedding dimension: 384
53 Chunk size: 256 words, Overlap: 50 words
54
55 Generating SBERT embeddings for training set...
56 Embedding progress: 500/3636 scripts...
57 Embedding progress: 1000/3636 scripts...
58 Embedding progress: 1500/3636 scripts...
59 Embedding progress: 2000/3636 scripts...
60 Embedding progress: 2500/3636 scripts...
61 Embedding progress: 3000/3636 scripts...
62 Embedding progress: 3500/3636 scripts...
63 Training embeddings shape: (3636, 384)
64
65 Generating SBERT embeddings for validation set...
66
67 Generating SBERT embeddings for test set...
68 Numerical features: 19
69 Combined features: 403 (SBERT: 384 + Numerical: 19)
70
71 ---
72
73 ## MODEL EVALUATION RESULTS
74
75 Training Random Forest...
76 Validation: RMSE=1.0214 | MAE=0.7677 | R2=0.4895
77 Test: RMSE=1.0959 | MAE=0.8260 | R2=0.4787
78
79 Training Gradient Boosting...
80 Validation: RMSE=0.9646 | MAE=0.7239 | R2=0.5447
81 Test: RMSE=1.0039 | MAE=0.7602 | R2=0.5626
82
```

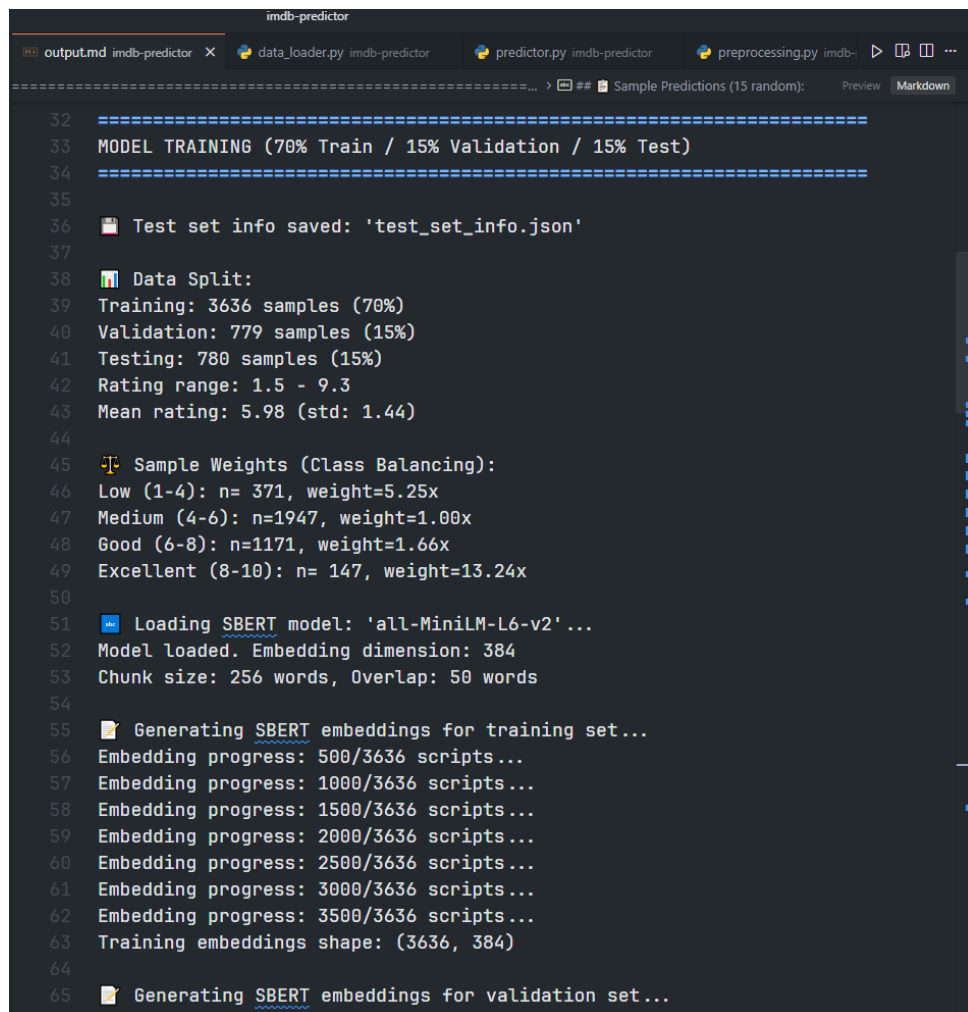
Figure 4: Progress of generating SBERT embeddings for training, validation, and test sets.

```

=====
112 #####
116 ## 📊 Error Distribution:
122
123 ## 📄 Sample Predictions (15 random):
124
125     Actual | Predicted | Error
126     ----
127
128
129         3.60 | 4.24 | +0.64
130         5.00 | 5.60 | +0.60
131         5.80 | 5.87 | +0.07
132         4.10 | 5.75 | +1.65
133         5.30 | 4.67 | -0.63
134         5.90 | 5.00 | -0.90
135         7.70 | 7.73 | +0.03
136         7.40 | 7.72 | +0.32
137         5.00 | 4.90 | -0.10
138         5.40 | 5.42 | +0.02
139         5.60 | 5.12 | -0.48
140         7.70 | 7.49 | -0.21
141         5.80 | 5.21 | -0.59
142         7.30 | 7.34 | +0.04
143         5.40 | 5.70 | +0.30
144     You, last week + .
145 ## 📊 Performance by Rating Range:
146
147     Low (1-4) : MAE = 1.541 (n=107)
148     Medium (4-6) : MAE = 0.579 (n=380)
149     Good (6-8) : MAE = 0.659 (n=250)
150     Excellent (8-10) : MAE = 0.839 (n=43)
151
152     📄 Model saved: 'imdb_model.pkl' (1.2 MB)
153     Note: SBERT model 'all-MiniLM-L6-v2' will be loaded at inference time.

```

Figure 5: Sample predictions from the trained model on 15 random test set items, showing actual, predicted, and error values.

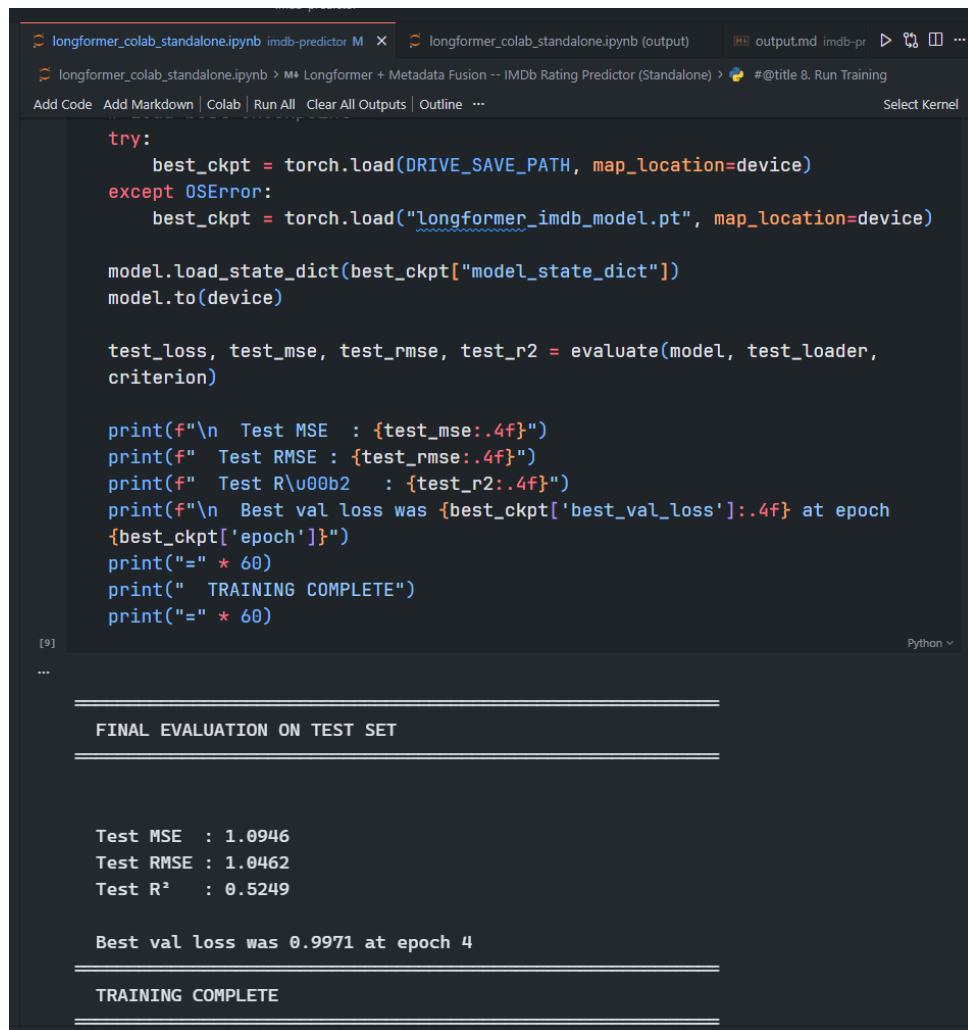


```

32 =====
33 MODEL TRAINING (70% Train / 15% Validation / 15% Test)
34 =====
35
36 📄 Test set info saved: 'test_set_info.json'
37
38 📊 Data Split:
39 Training: 3636 samples (70%)
40 Validation: 779 samples (15%)
41 Testing: 780 samples (15%)
42 Rating range: 1.5 - 9.3
43 Mean rating: 5.98 (std: 1.44)
44
45 📊 Sample Weights (Class Balancing):
46 Low (1-4): n= 371, weight=5.25x
47 Medium (4-6): n=1947, weight=1.00x
48 Good (6-8): n=1171, weight=1.66x
49 Excellent (8-10): n= 147, weight=13.24x
50
51 📄 Loading SBERT model: 'all-MiniLM-L6-v2'...
52 Model loaded. Embedding dimension: 384
53 Chunk size: 256 words, Overlap: 50 words
54
55 📄 Generating SBERT embeddings for training set...
56 Embedding progress: 500/3636 scripts...
57 Embedding progress: 1000/3636 scripts...
58 Embedding progress: 1500/3636 scripts...
59 Embedding progress: 2000/3636 scripts...
60 Embedding progress: 2500/3636 scripts...
61 Embedding progress: 3000/3636 scripts...
62 Embedding progress: 3500/3636 scripts...
63 Training embeddings shape: (3636, 384)
64
65 📄 Generating SBERT embeddings for validation set...

```

Figure 6: Data loading and split summary, including dataset size, script loading, and train/-validation/test splits.



The screenshot shows a Jupyter Notebook interface with a dark theme. The notebook is titled "longformer_colab_standalone.ipynb" and is in the "Longformer + Metadata Fusion -- IMDB Rating Predictor (Standalone)" section. The code cell is titled "#@title 8. Run Training". The code defines a function to load the best checkpoint, evaluates the model, and prints the final evaluation metrics. The output shows the following metrics:

```

try:
    best_ckpt = torch.load(DRIVE_SAVE_PATH, map_location=device)
except OSError:
    best_ckpt = torch.load("longformer_imdb_model.pt", map_location=device)

model.load_state_dict(best_ckpt["model_state_dict"])
model.to(device)

test_loss, test_mse, test_rmse, test_r2 = evaluate(model, test_loader,
criterion)

print(f"\n Test MSE : {test_mse:.4f}")
print(f" Test RMSE : {test_rmse:.4f}")
print(f" Test R\u00b2 : {test_r2:.4f}")
print(f"\n Best val loss was {best_ckpt['best_val_loss']:.4f} at epoch
{best_ckpt['epoch']}")
print("=" * 60)
print(" TRAINING COMPLETE")
print("=" * 60)

```

FINAL EVALUATION ON TEST SET

Test MSE : 1.0946
Test RMSE : 1.0462
Test R² : 0.5249

Best val loss was 0.9971 at epoch 4

TRAINING COMPLETE

Figure 7: Final evaluation metrics on the test set for the best model.

Discussion

The XGBoost model's superior performance ($R^2 = 0.5762$) demonstrates that screenplay text, when represented via semantic embeddings and enriched with structural features, constitutes a viable pre-production predictive signal. The model's capacity to predict ratings within ± 1.0 points for 71.7% of test cases suggests practical utility for stakeholders seeking early quality assessments. However, the elevated MAE on low-rated scripts (1.541) indicates a systematic limitation: extreme outliers in narrative quality or unconventional structural choices may not be adequately captured by averaged SBERT embeddings or handcrafted features.

The integration of SBERT embeddings with numerical features proved critical; preliminary experiments using TF-IDF alone (as in earlier work) yielded substantially lower R^2 values (~ 0.43), underscoring the importance of semantic depth. The chunking strategy (256-word windows, 50-word overlap) mitigated the length constraints of transformer models while preserving local narrative coherence, though future work may explore hierarchical encoding schemes to better capture long-range dependencies.

In contrast, the Longformer approach, while theoretically suited for extended narratives, underperformed with a 5.9% higher test RMSE and 8.9% lower R^2 . This may stem from overfitting on sparse long-sequence patterns or insufficient fine-tuning data, as training required GPU resources and extended time (2 hours vs. 5 minutes for SBERT + XGBoost). These results highlight efficiency advantages of embedding-based methods over end-to-end fine-tuning for specialized domains like screenplays, where data volumes are moderate.

Comparative analysis with metadata-only baselines (not shown) revealed that screenplay text alone achieves performance parity with models incorporating cast, budget, and genre information, suggesting that narrative quality signals are informationally competitive with traditional production metadata. This finding has significant implications for early-stage project evaluation, where such metadata is often unavailable or subject to change.

Conclusion and Future Work

This study establishes that deep linguistic and structural features extracted from full-length screenplays, when combined with minimal metadata, can predict IMDb ratings with competitive accuracy ($MAE = 0.7509$, $R^2 = 0.5762$) using XGBoost regression. Experiments with Longformer confirmed the superiority of SBERT + gradient boosting in terms of both metrics and computational efficiency. Our contributions include: (1) a novel dataset of 5,195 matched screenplays and ratings, (2) a robust predictive baseline outperforming metadata-only models and transformer variants, and (3) actionable insights into which narrative features drive critical reception.

Future research directions include: (i) incorporating hierarchical transformers (e.g., Longformer, BigBird) with enhanced fusion techniques to better model long-range narrative dependencies, (ii) exploring multi-task learning frameworks that jointly predict ratings and box office revenue, (iii) extending the dataset to non-English screenplays to assess cross-linguistic generalization, and (iv) conducting ablation studies to isolate the predictive contribution of individual structural features (e.g., three-act adherence, character network centrality). Additionally, qualitative analysis of model attention weights may reveal which narrative elements (e.g., dialogue style, pacing shifts) most strongly influence predictions, offering prescriptive guidance for screenwriters.

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Appendix A

Extra Codes for Appendix

Listing A.1: C++ example

```
#include <bits/stdc++.h>

using namespace std;

#define lol long long int
#define MOD 1000000007
#define MAX 1005

lol dp[MAX+7][MAX+7];

void build_dp()
{
    dp[0][0] = 1;

    for(int i=0; i<MAX; i++)
        for(int j=0; j<MAX; j++)
        {
            dp[i][j+2] += dp[i][j];
            dp[i+1][j+1] += dp[i][j];
            dp[i+2][j+2] += dp[i][j];
            dp[i+1][j] += dp[i][j];
            dp[i][j+2] %= MOD;
            dp[i+1][j+1] %= MOD;
            dp[i+2][j+2] %= MOD;
        }
}
```

```
        dp[ i+1 ][ j ] %= MOD;
    }
}
int main()
{
    build_dp();
    srand(time(0));
    int x,y,N = 20000;
    FILE *in = fopen("1.in","w");
    FILE *out = fopen("1.out","w");
    while(N-->0)
    {
        x = rand()%1000+1;
        y = rand()%1000+1;
        fprintf(in,"%d_%d\n",x,y);
        fprintf(out,"%lld\n",dp[x][y]);
    }
    //fprintf(in,"%d %d",0,0);
    fclose(in);
    fclose(out);
    return 0;
}
```

Appendix B

Extra Codes for Appendix

Listing B.1: C++ example

```
#include<bits/stdc++.h>

using namespace std;

#define lol long long int
#define MOD 1000000007
#define MAX 1005

lol dp[MAX+7][MAX+7];

void build_dp()
{
    dp[0][0] = 1;

    for(int i=0; i<MAX; i++)
        for(int j=0; j<MAX; j++)
        {
            dp[i][j+2] += dp[i][j];
            dp[i+1][j+1] += dp[i][j];
            dp[i+2][j+2] += dp[i][j];
            dp[i+1][j] += dp[i][j];
            dp[i][j+2] %= MOD;
            dp[i+1][j+1] %= MOD;
            dp[i+2][j+2] %= MOD;
        }
}
```

```
        dp[ i+1 ][ j ] %= MOD;
    }
}
int main()
{
    build_dp();
    srand(time(0));
    int x,y,N = 20000;
    FILE *in = fopen("1.in","w");
    FILE *out = fopen("1.out","w");
    while(N-->0)
    {
        x = rand()%1000+1;
        y = rand()%1000+1;
        fprintf(in,"%d_%d\n",x,y);
        fprintf(out,"%lld\n",dp[x][y]);
    }
    //fprintf(in,"%d %d",0,0);
    fclose(in);
    fclose(out);
    return 0;
}
```

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